Marmara University, Engineering Faculty
Electrical and Electronics Engineering
EE4078 Fundamentals of Computer Networks
Final Exam Solutions, FALL2024

1.
a. The transmission at $t = 2$ was L-->G, so the answer is G,L. (7)
b. The transmission at $t = 1$ was K-->B, so the answer is n/a. (6)
c. The transmission at $t = 3$ was A-->B, so the answer is n/a. (6)
d. The transmission at $t = 5$ was J-->A, so the answer is A,J (6)

2.
a. At point 6,
source mac address is 78-A3-C8-35-42-C7, (5)
destination mac address is 1D-21-6C-13-CD-90, (5)
source IP address is 128.119.250.143 (5)
destination IP address is 128.119.250.55. (5)
b. No, datagrams can be sent across the subnet via the link layer in one go (5)

3.
a. The first step corresponds to C, second step corresponds to G, third step corresponds to B, fourth step corresponds to D, fifth step corresponds to A, sixth step corresponds to E, seventh and final step corresponds to F. C (7), G (3), B (3), D (3), A (3), E (3), F (3)

4. After you generate a pre-master secret (PMS), you will encrypt it with your friend's public key and then send the encrypted PMS to Trudy. Trudy will not be able to decrypt the PMS, since Trudy does not have your friend's private key. Thus, Trudy will not be able to determine the shared authentication key. Trudy may instead guess one by choosing a random key. During the last step of the handshake, Trudy sends to you a message authentication code (MAC) of all the handshake messages, using the guessed authentication key. When you receive the MAC, the MAC test will fail, and you will end the TCP connection. (5)